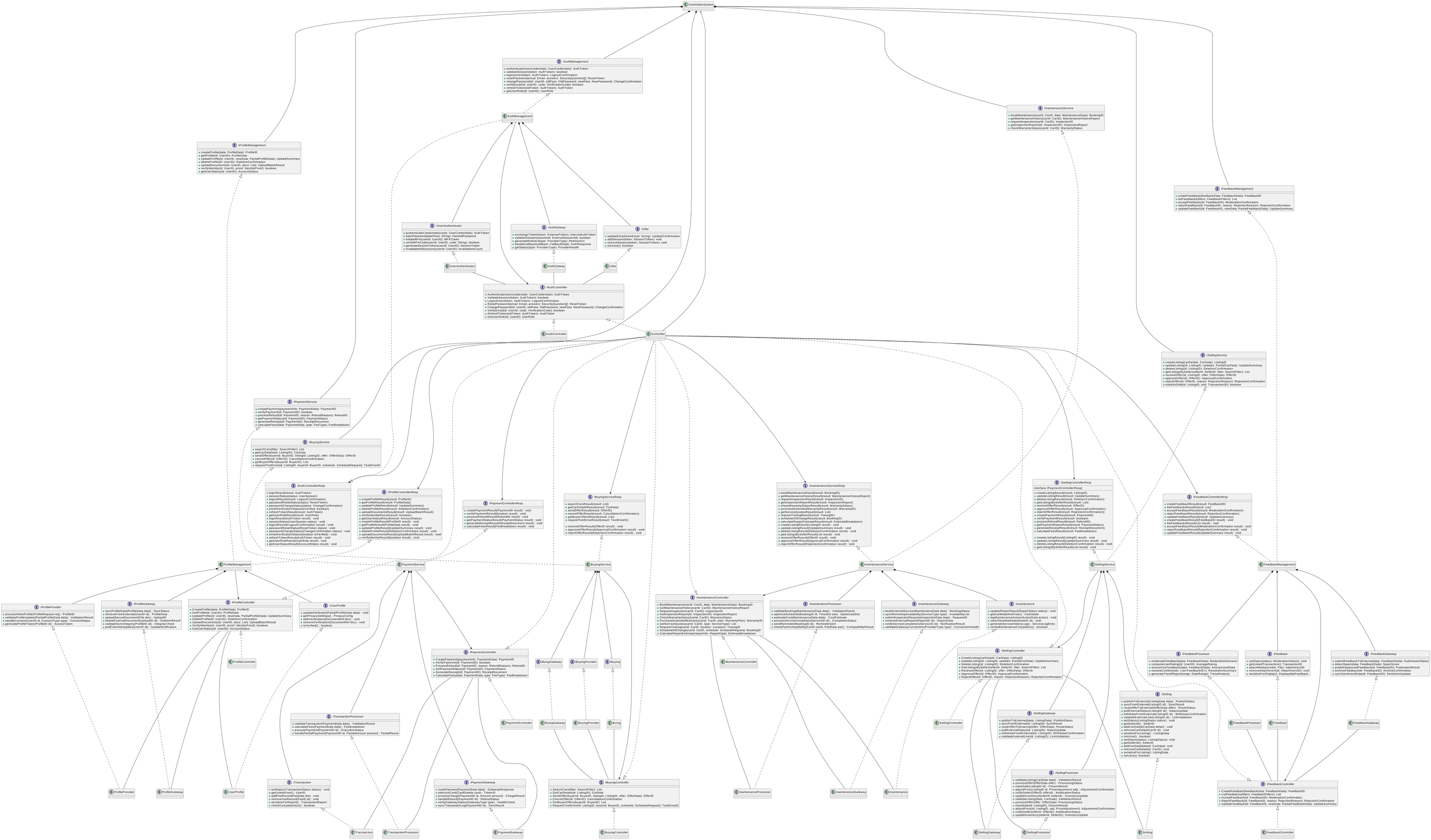

CareSallesSystem
IAuthManagement
+ authenticateUser(credentials: UserCredentials): AuthToken
+ validateSession(token: AuthToken): boolean
+ logoutUser(token: AuthToken): LogoutConfirmation
+ resetPassword(email: Email, answers: SecurityQuestion[]): ResetToken
+ changePassword(id: UserID, oldPass: OldPassword, newPass: NewPassword): ChangeConfirmation
+ verifyEmail(id: UserID, code: VerificationCode): boolean
+ refreshToken(oldToken: AuthToken): AuthToken
+ getUserRole(id: UserID): UserRole
AuthManagement
ImaintenanceService
+ bookMaintenance(carId: CarID, data: MaintenanceData): BookingID
+ getMaintenanceHistory(carId: CarID): MaintenanceHistoryReport
+ requestInspection(carId: CarID): InspectionID
+ getInspectionReport(id: InspectionID): InspectionReport
+ checkWarrantyStatus(carId: CarID): WarrantyStatus
IProfileManagement
+ createProfile(data: ProfileData): ProfileID
+ getProfile(id: UserID): ProfileData
+ updateProfile(id: UserID, newData: PartialProfileData): UpdateSummary
+ deleteProfile(id: UserID): DeletionConfirmation
+ uploadDocuments(id: UserID, docs: List): UploadBatchResult
+ verifyIdentity(id: UserID, proof: IdentityProof): boolean
+ getUserStatus(id: UserID): AccountStatus
IFeedbackManagement
+ createFeedback(feedbackData: FeedbackData): FeedbackID
+ listFeedbacks(filters: FeedbackFilters): List
+ acceptFeedback(id: FeedbackID): ModerationConfirmation
+ rejectFeedback(id: FeedbackID, reason: RejectionReason): RejectionConfirmation
+ updateFeedback(id: FeedbackID, newData: PartialFeedbackData): UpdateSummary
IUserAuthenticator
+ authenticateCredentials(creds: UserCredentials): AuthToken
+ hashPassword(plainPass: String): HashedPassword
+ initiateMFA(userId: UserID): MFAToken
+ verifyMFACode(userId: UserID, code: String): boolean
+ generateSessionToken(userId: UserID): SessionToken
+ invalidateAllSessions(userId: UserID): InvalidationCount
IAuthGatway
+ exchangeToken(token: ExternalToken): InternalAuthToken
+ validateSession(sessionId: ExternalSessionId): boolean
+ generateRedirect(type: ProviderType): RedirectUrl
+ handleCallback(callback: CallbackData): AuthResponse
+ getStatus(type: ProviderType): ProviderHealth
IUSer
+ updateEmail(newEmail: String): UpdateConfirmation
+ addSession(token: SessionToken): void
+ removeSession(token: SessionToken): void
+ isActive(): boolean
UserAuthenticator
AuthGatway
USer
IAuthController
+ AuthenticateUser(credentials: UserCredentials): AuthToken
+ ValidateSession(token: AuthToken): boolean
+ LogoutUser(token: AuthToken): LogoutConfirmation
+ ResetPassword(email: Email, answers: SecurityQuestion[]): ResetToken
+ ChangePassword(id: UserID, oldPass: OldPassword, newPass: NewPassword): ChangeConfirmation
+ VerifyEmail(id: UserID, code: VerificationCode): boolean
+ RefreshToken(oldToken: AuthToken): AuthToken
+ GetUserRole(id: UserID): UserRole
AuthController
Controller
ISellingService
+ createListing(carDetails: CarData): ListingID
+ updateListing(id: ListingID, updates: PartialCarData): UpdateSummary
+ deleteListing(id: ListingID): DeletionConfirmation
+ getListingsBySeller(sellerId: SellerID, filter: SearchFilter): List
+ receiveOffer(id: ListingID, offer: OfferData): OfferID
+ approveOffer(id: OfferID): ApprovalConfirmation
+ rejectOffer(id: OfferID, reason: RejectionReason): RejectionConfirmation
+ markAsSold(id: ListingID, txId: TransactionID): boolean
IPaymentService
+ createPayment(paymentInfo: PaymentData): PaymentID
+ verifyPayment(id: PaymentID): boolean
+ processRefund(id: PaymentID, reason: RefundReason): RefundID
+ getPaymentStatus(id: PaymentID): PaymentStatus
+ generateReceipt(id: PaymentID): ReceiptDocument
+ calculateFees(data: PaymentData, type: FeeType): FeeBreakdown
IBuyingService
+ searchCars(filter: SearchFilter): List
+ getCarDetails(id: ListingID): CarData
+ sendOffer(buyerId: BuyerID, listingId: ListingID, offer: OfferData): OfferID
+ cancelOffer(id: OfferID): CancellationConfirmation
+ getBuyerOffers(buyerId: BuyerID): List
+ requestTestDrive(id: ListingID, buyerId: BuyerID, schedule: ScheduleRequest): TestDriveID
IAuthControllerResp
+ loginResult(result: AuthToken)
+ sessionStatus(status: UserSession)
+ logoutResult(result: LogoutConfirmation)
+ passwordResetStatus(status: ResetToken)
+ passwordChangeStatus(status: ChangeConfirmation)
+ emailVerificationStatus(isVerified: boolean)
+ refreshTokenResult(result: AuthToken)
+ getUserRoleResult(result: UserRole)
+ loginResult(AuthToken result) : void
+ sessionStatus(UserSession status) : void
+ logoutResult(LogoutConfirmation result) : void
+ passwordResetStatus(ResetToken status) : void
+ passwordChangeStatus(ChangeConfirmation status) : void
+ emailVerificationStatus(boolean isVerified) : void
+ refreshTokenResult(AuthToken result) : void
+ getUserRoleResult(UserRole result) : void
+ getUserStatusResult(AccountStatus result) : void
IProfileControllerResp
+ createProfileResult(result: ProfileID)
+ getProfileResult(result: ProfileData)
+ updateProfileResult(result: UpdateSummary)
+ deleteProfileResult(result: DeletionConfirmation)
+ uploadDocumentsResult(result: UploadBatchResult)
+ verifyIdentityResult(result: boolean)
+ getUserStatusResult(result: AccountStatus)
+ createProfileResult(ProfileID result) : void
+ getProfileResult(ProfileData result) : void
+ updateProfileResult(UpdateSummary result) : void
+ deleteProfileResult(DeletionConfirmation result) : void
+ uploadDocumentsResult(UploadBatchResult result) : void
+ verifyIdentityResult(boolean result) : void
IPaymentControllerResp
+ createPaymentResult(PaymentID result) : void
+ verifyPaymentResult(boolean result) : void
+ processRefundResult(RefundID result) : void
+ getPaymentStatusResult(PaymentStatus result) : void
+ generateReceiptResult(ReceiptDocument result) : void
+ calculateFeesResult(FeeBreakdown result) : void
IBuyingServiceResp
+ searchCarsResult(result: List)
+ getCarDetailsResult(result: CarData)
+ sendOfferResult(result: OfferID)
+ cancelOfferResult(result: CancellationConfirmation)
+ getBuyerOffersResult(result: List)
+ requestTestDriveResult(result: TestDriveID)
+ receiveOfferResult(OfferID result) : void
+ approveOfferResult(ApprovalConfirmation result) : void
+ rejectOfferResult(RejectionConfirmation result) : void
ImaintenanceServiceResp
+ bookMaintenanceResult(result: BookingID)
+ getMaintenanceHistoryResult(result: MaintenanceHistoryReport)
+ requestInspectionResult(result: InspectionID)
+ getInspectionReportResult(result: InspectionReport)
+ checkWarrantyStatusResult(result: WarrantyStatus)
+ purchaseExtendedWarrantyResult(result: WarrantyID)
+ getServiceQuotesResult(result: List)
+ requestTowingResult(result: TowingID)
+ scheduleOilChangeResult(result: BookingID)
+ calculateRepairEstimateResult(result: EstimateBreakdown)
+ createListingResult(ListingID result) : void
+ updateListingResult(UpdateSummary result) : void
+ deleteListingResult(DeletionConfirmation result) : void
+ getListingsBySellerResult(List result) : void
+ receiveOfferResult(OfferID result) : void
+ approveOfferResult(ApprovalConfirmation result) : void
+ rejectOfferResult(RejectionConfirmation result) : void
ISellingControllerResp
interface IPaymentControllerResp{
+ createListingResult(result: ListingID)
+ updateListingResult(result: UpdateSummary)
+ deleteListingResult(result: DeletionConfirmation)
+ getListingsBySellerResult(result: List)
+ receiveOfferResult(result: OfferID)
+ approveOfferResult(result: ApprovalConfirmation)
+ rejectOfferResult(result: RejectionConfirmation)}
+ createPaymentResult(result: PaymentID)
+ verifyPaymentResult(result: boolean)
+ processRefundResult(result: RefundID)
+ getPaymentStatusResult(result: PaymentStatus)
+ generateReceiptResult(result: ReceiptDocument)
+ calculateFeesResult(result: FeeBreakdown)
+ createListingResult(ListingID result) : void
+ updateListingResult(UpdateSummary result) : void
+ deleteListingResult(DeletionConfirmation result) : void
+ getListingsBySellerResult(List result) : void
IFeedbackControllerResp
+ createFeedbackResult(result: FeedbackID)
+ listFeedbacksResult(result: List)
+ acceptFeedbackResult(result: ModerationConfirmation)
+ rejectFeedbackResult(result: RejectionConfirmation)
+ updateFeedbackResult(result: UpdateSummary)
+ createFeedbackResult(FeedbackID result) : void
+ listFeedbacksResult(List result) : void
+ acceptFeedbackResult(ModerationConfirmation result) : void
+ rejectFeedbackResult(RejectionConfirmation result) : void
+ updateFeedbackResult(UpdateSummary result) : void
ProfileManagement
PaymentService
BuyingService
maintenanceService
SellingService
FeedbackManagement
IProfileProvider
+ provisionNewProfile(ProfileRequest req) : ProfileID
+ validateProfileUpdate(PartialProfileData data) : ValidationResult
+ handleConsent(UserID id, ConsentType type) : ConsentStatus
+ generateProfileToken(ProfileID id) : AccessToken
IProfileGateway
+ syncProfileData(ProfileData data) : SyncStatus
+ retrieveFromExternal(UserID id) : ProfileData
+ uploadSecureDocument(File doc) : UploadID
+ deleteExternalDocument(UploadID id) : DeletionResult
+ validateSyncIntegrity(ProfileID id) : IntegrityCheck
+ pollExternalUpdates(UserID id) : UpdateNotification
IProfileController
+ CreateProfile(data: ProfileData): ProfileID
+ GetProfile(id: UserID): ProfileData
+ UpdateProfile(id: UserID, newData: PartialProfileData): UpdateSummary
+ DeleteProfile(id: UserID): DeletionConfirmation
+ UploadDocuments(id: UserID, docs: List): UploadBatchResult
+ VerifyIdentity(id: UserID, proof: IdentityProof): boolean
+ GetUserStatus(id: UserID): AccountStatus
IUserProfile
+ updateAttributes(PartialProfileData data) : void
+ getPrivacySettings() : PrivacyConfig
+ addVerification(DocumentRef doc) : void
+ removeVerification(DocumentRef doc) : void
+ isVerified() : boolean
ProfileController
ImaintenanceController
+ BookMaintenance(carId: CarID, data: MaintenanceData): BookingID
+ GetMaintenanceHistory(carId: CarID): MaintenanceHistoryReport
+ RequestInspection(carId: CarID): InspectionID
+ GetInspectionReport(id: InspectionID): InspectionReport
+ CheckWarrantyStatus(carId: CarID): WarrantyStatus
+ PurchaseExtendedWarranty(carId: CarID, plan: WarrantyPlan): WarrantyID
+ GetServiceQuotes(carId: CarID, type: ServiceType): List
+ RequestTowing(carId: CarID, location: Location): TowingID
+ ScheduleOilChange(carId: CarID, schedule: ScheduleRequest): BookingID
+ CalculateRepairEstimate(repairInfo: RepairData): EstimateBreakdown
ImaintenanceProcessor
+ validateBooking(MaintenanceData data) : ValidationResult
+ optimizeSchedule(BookingID id, TimeSlot slot) : OptimizedSlot
+ estimateCost(MaintenanceData data) : CostEstimate
+ processServiceCompletion(ServiceID id) : CompletionStatus
+ sendReminder(BookingID id) : ReminderSent
+ checkPartCompatibility(CarID carId, PartData part) : CompatibilityResult
ImaintenanceGateway
+ bookExternalService(MaintenanceData data) : BookingStatus
+ syncWorkshopAvailability(ServiceType type) : AvailabilityList
+ submitInspectionRequest(InspectionData data) : RequestID
+ retrieveExternalReport(ReportID id) : ReportData
+ notifyServiceCompletion(ServiceID id) : NotificationResult
+ validateGatewayConnection(ProviderType type) : ConnectionHealth
Imaintenance
+ updateRepairStatus(RepairStatus status) : void
+ getCarModelAndYear() : CarDetails
+ logMaintenanceAction(ActionData action) : void
+ clearObsoleteNotes(NoteID id) : void
+ generateServiceHistoryLog() : ServiceLogEntry
+ verifyMaintenanceCompletion() : boolean
MaintenanceController
IPaymentController
+ CreatePayment(paymentInfo: PaymentData): PaymentID
+ VerifyPayment(id: PaymentID): boolean
+ ProcessRefund(id: PaymentID, reason: RefundReason): RefundID
+ GetPaymentStatus(id: PaymentID): PaymentStatus
+ GenerateReceipt(id: PaymentID): ReceiptDocument
+ CalculateFees(data: PaymentData, type: FeeType): FeeBreakdown
IBuingGateway
IBuyingProvidor
IBuying
ISellingController
+ CreateListing(carDetails: CarData): ListingID
+ UpdateListing(id: ListingID, updates: PartialCarData): UpdateSummary
+ DeleteListing(id: ListingID): DeletionConfirmation
+ GetListingsBySeller(sellerId: SellerID, filter: SearchFilter): List
+ ReceiveOffer(id: ListingID, offer: OfferData): OfferID
+ ApproveOffer(id: OfferID): ApprovalConfirmation
+ RejectOffer(id: OfferID, reason: RejectionReason): RejectionConfirmation
IFeedbackProcessor
+ moderateFeedback(data: FeedbackData): ModerationDecision
+ computeUserRating(id: UserID): AverageRating
+ anonymizeFeedback(data: FeedbackData): AnonymizedData
+ resolveConflicts(ids: List<FeedbackID>): ResolutionSummary
+ generateTrendReport(range: DateRange): TrendAnalysis
IFeedback
+ setStatus(status: ModerationStatus): void
+ getLinkedTransaction(): TransactionID
+ attachMedia(media: File): AttachmentID
+ removeAttachment(id: AttachmentID): void
+ serializeForDisplay(): DisplayableFeedback
IFeedbackGateway
+ submitFeedbackToExternal(data: FeedbackData): SubmissionStatus
+ detectSpam(data: FeedbackData): SpamScore
+ publishApprovedFeedback(id: FeedbackID): PublicationResult
+ archiveFeedback(id: FeedbackID): ArchiveConfirmation
+ syncSentimentData(id: FeedbackID): SentimentUpdate
ISelling
+ publishToExternal(ListingData data) : PublishStatus
+ syncFromExternal(ListingID id) : SyncResult
+ routeOfferToExternal(OfferData offer) : RouteStatus
+ pollExternalStatus(ListingID id) : StatusUpdate
+ withdrawFromExternal(ListingID id) : WithdrawConfirmation
+ validateExternalLink(ListingID id) : LinkValidation
+ setStatus(ListingStatus status) : void
+ getSellerId() : SellerID
+ addCarDetail(CarData detail) : void
+ removeCarDetail(CarID id) : void
+ serializeForListing() : ListingData
+ isActive() : boolean
+ setStatus(status: ListingStatus): void
+ getSellerId(): SellerID
+ addCarDetail(detail: CarData): void
+ removeCarDetail(id: CarID): void
+ serializeForListing(): ListingData
+ isActive(): boolean
ITransiactionProcessor
+ validateTransaction(PaymentData data) : ValidationResult
+ calculateFees(PaymentData data) : FeeBreakdown
+ executePayment(PaymentID id) : ExecutionStatus
+ handlePartialPayment(PaymentID id, PartialAmount amount) : PartialResult
PaymentController
BuingGateway
BuyingProvidor
Buing
SellingController
ISellingGateway
+ publishToExternal(data: ListingData): PublishStatus
+ syncFromExternal(id: ListingID): SyncResult
+ routeOfferToExternal(offer: OfferData): RouteStatus
+ pollExternalStatus(id: ListingID): StatusUpdate
+ withdrawFromExternal(id: ListingID): WithdrawConfirmation
+ validateExternalLink(id: ListingID): LinkValidation
FeedbackProcessor
Feedback
FeedbackGateway
ITransiaction
+ setStatus(TransactionStatus status) : void
+ getLinkedUser() : UserID
+ addFeeRecord(FeeData fee) : void
+ removeFeeRecord(FeeID id) : void
+ serializeForReport() : TransactionReport
+ checkCompleteness() : boolean
IPaymentGateway
+ routePayment(PaymentData data) : GatewayResponse
+ tokenizeCard(CardDetails card) : TokenID
+ processCharge(PaymentID id, Amount amount) : ChargeResult
+ handleRefund(PaymentID id) : RefundStatus
+ verifyGatewayStatus(GatewayType type) : HealthCheck
+ syncTransactionLog(PaymentID id) : SyncResult
IBuyingController
+ SearchCars(filter: SearchFilter): List
+ GetCarDetails(id: ListingID): CarData
+ SendOffer(buyerId: BuyerID, listingId: ListingID, offer: OfferData): OfferID
+ CancelOffer(id: OfferID): CancellationConfirmation
+ GetBuyerOffers(buyerId: BuyerID): List
+ RequestTestDrive(id: ListingID, buyerId: BuyerID, schedule: ScheduleRequest): TestDriveID
ISellingProcessor
+ validateListing(CarData data) : ValidationResult
+ processOffer(OfferData offer) : ProcessingStatus
+ closeSale(ListingID id) : ClosureResult
+ adjustPrice(ListingID id, PriceAdjustment adj) : AdjustmentConfirmation
+ notifySeller(OfferID offerId) : NotificationStatus
+ updateInventory(SellerID sellerId) : InventoryUpdate
+ validateListing(data: CarData): ValidationResult
+ processOffer(offer: OfferData): ProcessingStatus
+ closeSale(id: ListingID): ClosureResult
+ adjustPrice(id: ListingID, adj: PriceAdjustment): AdjustmentConfirmation
+ notifySeller(offerId: OfferID): NotificationStatus
+ updateInventory(sellerId: SellerID): InventoryUpdate
IFeedbackController
+ CreateFeedback(feedbackData: FeedbackData): FeedbackID
+ ListFeedbacks(filters: FeedbackFilters): List
+ AcceptFeedback(id: FeedbackID): ModerationConfirmation
+ RejectFeedback(id: FeedbackID, reason: RejectionReason): RejectionConfirmation
+ UpdateFeedback(id: FeedbackID, newData: PartialFeedbackData): UpdateSummary
ProfileProvider
ProfileGateway
UserProfile
maintenanceProcessor
maintenanceGateway
maintenance
Transiaction
TransiactionProcessor
PaymentGateway
BuyingController
SellingGateway
SellingProcessor
Selling
FeedbackController